# Topic 3: Solved Problems

Full step-by-step analytical resolutions for the 6 official problem set exercises on Diffusion in Solids and Mass Transport, formatted with rigorous 4-phase pedagogical derivations.

**OFFICIAL UC3M ERROR FUNCTION REFERENCE TABLE [SESSION 5 SLIDE 21 ]**

| $z$ | $\text{erf}(z)$ | $z$ | $\text{erf}(z)$ | $z$ | $\text{erf}(z)$ | $z$ | $\text{erf}(z)$ |
|---|---|---|---|---|---|---|---|
| 0.00 | 0.0000 | 0.25 | 0.2763 | 0.70 | 0.6778 | 1.30 | 0.9340 |
| 0.01 | 0.0113 | 0.30 | 0.3286 | 0.75 | 0.7112 | 1.40 | 0.9523 |
| 0.02 | 0.0226 | 0.35 | 0.3794 | 0.80 | 0.7421 | 1.50 | 0.9661 |
| 0.03 | 0.0338 | 0.40 | 0.4284 | 0.85 | 0.7707 | 1.60 | 0.9763 |
| 0.04 | 0.0451 | 0.45 | 0.4755 | 0.90 | 0.7969 | 1.70 | 0.9838 |
| 0.05 | 0.0564 | 0.50 | 0.5205 | 0.95 | 0.8209 | 1.80 | 0.9891 |
| 0.10 | 0.1125 | 0.55 | 0.5633 | 1.00 | 0.8427 | 1.90 | 0.9928 |
| 0.15 | 0.1680 | 0.60 | 0.6039 | 1.10 | 0.8802 | 2.00 | 0.9953 |
| 0.20 | 0.2227 | 0.65 | 0.6420 | 1.20 | 0.9103 | | |

`Problem 1` `Problems T3_Diffusion.pdf #1` `Fick's 2nd Law · erf(z)`

## Cementation of Grade 1018 Steel Gear

`Original statement · Problems T3_Diffusion.pdf`

*1. In a steel gear of grade 1018 (*$0.18$ wt% C*) a cementation process is carried out at* $927^\circ\text{C}$*. Calculate the time required in order to increase the* C *content up to* $0.3$ wt% *at* $0.6$ mm *below the surface of the gear. Assume that the* C *content in the surface is* $1$ wt%*. Data:* $D_{\text{C in } \gamma\text{-Fe}} = 1.28 \times 10^{-11}\ \text{m}^2/\text{s}$.

`Official Solution: t = 6636 s (1.84 h)`

PHASE 1 · HYPOTHESES & PARAMETERS

- **Semi-Infinite Medium:** The penetration depth $x = 0.6$ mm is small compared with the size of the part, allowing one-dimensional semi-infinite modeling [Session 5 Slide 20].
- **Constant Surface Concentration:** The carburizing atmosphere maintains $C(0,t) = C_s = 1.0$ wt% for $t > 0$.
- **Homogeneous Initial State:** The AISI 1018 steel core has initial carbon concentration $C(x,0) = C_0 = 0.18$ wt%.
- **Parameters:** $C_0 = 0.18$ wt%, $C_s = 1.00$ wt%, $C_x = 0.30$ wt%, $x = 0.6$ mm $= 6.0 \times 10^{-4}$ m, $D = 1.28 \times 10^{-11}$ m$^2$/s at $T = 927^\circ$C.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

In non-steady-state diffusion into a semi-infinite solid with constant surface composition, Fick's Second Law yields [Session 5 Slide 20]:

$$\frac{C_s - C_x}{C_s - C_0} = \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right) = \operatorname{erf}(z)$$

We evaluate the concentration ratio, locate the corresponding argument $z$ in the official error function table via exact linear interpolation, and solve for time $t$:

$$z = \frac{x}{2\sqrt{Dt}} \implies t = \frac{x^2}{4 \cdot z^2 \cdot D}$$

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

**1. Concentration Ratio:**

$$\frac{C_s - C_x}{C_s - C_0} = \frac{1.00 - 0.30}{1.00 - 0.18} = \frac{0.70}{0.82} \approx 0.8536585 \implies \operatorname{erf}(z) = 0.8536585$$

**2. Linear Interpolation in Error Function Table [Slide 21]:**

- For $z_1 = 1.00 \implies \operatorname{erf}(z_1) = 0.8427$
- For $z_2 = 1.10 \implies \operatorname{erf}(z_2) = 0.8802$

$$\frac{z-1.00}{1.10-1.00}=\frac{0.8536585-0.8427}{0.8802-0.8427}\implies\frac{z-1.00}{0.10}=\frac{0.0109585}{0.0375}\approx 0.292227$$

$$z=1.00+0.10\times 0.292227=1.02922\implies z^2=(1.02922)^2\approx 1.05930$$

**3. Time of Carburization:**

$$x^2=(6.0\times10^{-4}\ \mathrm{m})^2=3.60\times10^{-7}\ \mathrm{m}^2$$

$$4\cdot z^2\cdot D=4\times1.05930\times(1.28\times10^{-11}\ \mathrm{m}^2/\mathrm{s})=5.4236\times10^{-11}\ \mathrm{m}^2/\mathrm{s}$$

$$t=\frac{3.60\times10^{-7}\ \mathrm{m}^2}{5.4236\times10^{-11}\ \mathrm{m}^2/\mathrm{s}}\approx\mathbf{6638}\ \mathrm{s}=\mathbf{1.844}\ \mathrm{h}$$

**Discrepancy with the official solution.** The official key gives $t=6636$ s. Linear interpolation in the table gives $z=1.0292$ and $t=6637.6$ s $\approx 6638$ s (the exact inverse error function, $z=1.0271$, would give $6665$ s because the table is linear between $1.00$ and $1.10$). The key is $0.02\%$ below the interpolated value; this is a rounding difference, and the interpolated value is reported.

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

**Dimensional Check:** $[t]=\frac{\mathrm{m}^2}{\mathrm{m}^2/\mathrm{s}}=\mathrm{s}$. **Physical Insight:** About $1.84$ hours at $927^\circ\mathrm{C}$ bring the carbon content at $0.6$ mm to $0.30$ wt$\%$. Carbon is small, so it diffuses interstitially [Slide 13]; in the surface layer it blocks the movement of dislocations and hardens the steel for wear resistance, while the core stays tough, and the excess carbon leaves residual compressive stresses [Slide 30].

Problem 2 | Problems T3_Diffusion.pdf #2 | Arrhenius · Semiconductor Doping

# Diffusion of Aluminum in Monocrystalline Silicon

Original statement · Problems T3_Diffusion.pdf

*2.-* Al *can diffuse in a* Si *monocrystal. Calculate the temperature at which the diffusion coefficient will have a value of* $10^{-14}\ \mathrm{m^2/s}$*? Data:* $E_D = 73\ \mathrm{kcal/mol}$*;* $D_0 = 1.55 \times 10^{-4}\ \mathrm{m^2/s}$ *and* $R = 1.987\ \mathrm{cal/mol \cdot K}$. Official Solution: T = 1566 K (1293 °C)

PHASE 1 · HYPOTHESES & PARAMETERS

- **Volume Diffusion in a Single Crystal:** A single crystal has no grain boundaries, so diffusion takes place through the volume, the slowest path ($D_{surface} > D_{grain\ boundary} > D_{volume}$) [Slide 25].
- **Arrhenius Behavior:** The temperature dependence of $D$ obeys Arrhenius kinetics with constant $D_0$ and $E_D$ [Slide 26].
- **Parameters:** $D = 10^{-14}\ \mathrm{m^2/s}$, $D_0 = 1.55 \times 10^{-4}\ \mathrm{m^2/s}$, $E_D = 73\,000\ \mathrm{cal/mol}$, $R = 1.987\ \mathrm{cal/mol \cdot K}$.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

Starting from the Arrhenius equation [Session 5 Slide 26]:

$$D = D_0 \exp\left(-\frac{E_D}{RT}\right) \implies \ln\left(\frac{D_0}{D}\right) = \frac{E_D}{RT}$$

Isolating absolute temperature $T$:

$$T = \frac{E_D}{R \cdot \ln(D_0/D)}$$

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

**1. Diffusivity Ratio:**

$$\frac{D_0}{D} = \frac{1.55 \times 10^{-4}\ \mathrm{m^2/s}}{1.0 \times 10^{-14}\ \mathrm{m^2/s}} = 1.55 \times 10^{10}$$

**2. Natural Logarithm:**

$$\ln(1.55 \times 10^{10}) = \ln(1.55) + 10 \cdot \ln(10) = 0.438255 + 23.025851 = 23.464106$$

**3. Absolute Temperature Calculation:**

$$T = \frac{73\,000\ \mathrm{cal/mol}}{(1.987\ \mathrm{cal/mol \cdot K}) \times (23.464106)} = \frac{73\,000}{46.62318} \approx \mathbf{1565.74}\ \mathrm{K} \approx \mathbf{1566}\ \mathrm{K}$$

**Conversion to Celsius:**

$$T = 1565.74 - 273.15 = \mathbf{1292.59}^\circ\mathrm{C} \approx \mathbf{1293}^\circ\mathrm{C}$$

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

**Dimensional Check:** $[T] = \frac{\mathrm{cal/mol}}{(\mathrm{cal/mol \cdot K})} = \mathrm{K}$. **Physical Insight:** A diffusivity of $10^{-14}\ \mathrm{m^2/s}$ needs $1293^\circ\mathrm{C}$: $D$ depends exponentially on $T$, and the high activation energy ($73\ \mathrm{kcal/mol} \approx 305\ \mathrm{kJ/mol}$) means the atoms need a lot of thermal energy to jump [Slides 4, 26]. Diffusion of impurities in Si wafers is the process used to fabricate integrated circuits [Slide 29].

Problem 3 | Problems T3_Diffusion.pdf #3 | Two-Point Arrhenius · Activation Energy

# Activation Energy and Diffusivity of Carbon in Mild Steel

Original statement · Problems T3_Diffusion.pdf

*3.- The diffusivity of carbon in mild steel was measured at two different temperatures:*
*Data:* $R = 8.314\ \text{J/mol} \cdot \text{K}$
• *At* $T_1 = 850^\circ\text{C}$*:* $D_1 = 4.826 \times 10^{-12}\ \text{m}^2/\text{s}$
• *At* $T_2 = 950^\circ\text{C}$*:* $D_2 = 1.805 \times 10^{-11}\ \text{m}^2/\text{s}$
*Using these data, calculate:*
*a) The activation energy for the* C *diffusion in mild steel, for the temperature interval of* $850^\circ\text{C}$ *to* $950^\circ\text{C}$ *(Solution:* $E_D = 36\ \text{kcal/mol}$*)*
*b) The diffusivity at* $1000^\circ\text{C}$*, assuming that the activation energy does not change at this temperature range. (Solution:* $D = 3.23 \times 10^{-11}\ \text{m}^2/\text{s}$*)*

Official Solution: ED = 36 kcal/mol; D(1000 °C) = 3.23 x 10^-11 m^2/s

PHASE 1 · HYPOTHESES & PARAMETERS

- **Single Arrhenius Law (assumption):** The problem uses one Arrhenius law with constant $D_0$ and $E_D$ for C in mild steel from $850^\circ\text{C}$ to $1000^\circ\text{C}$, as part b) states [Session 5 Slide 26].
- **Parameters:** $T_1 = 850^\circ\text{C} = 1123.15\ \text{K}$, $D_1 = 4.826 \times 10^{-12}\ \text{m}^2/\text{s}$ $T_2 = 950^\circ\text{C} = 1223.15\ \text{K}$, $D_2 = 1.805 \times 10^{-11}\ \text{m}^2/\text{s}$ $T_3 = 1000^\circ\text{C} = 1273.15\ \text{K}$, $R = 8.314\ \text{J/mol} \cdot \text{K}$.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

Dividing Arrhenius equations at two temperatures eliminates $D_0$ [Session 5 Slide 26]:

$$\ln\left(\frac{D_2}{D_1}\right) = \frac{E_D}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) \implies E_D = \frac{R \cdot \ln(D_2/D_1)}{\frac{1}{T_1} - \frac{1}{T_2}}$$

For part (b), extrapolating to $T_3 = 1000^\circ\text{C}$:

$$D_3 = D_2 \cdot \exp\left[\frac{E_D}{R}\left(\frac{1}{T_2} - \frac{1}{T_3}\right)\right]$$

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

**Part (a): Activation Energy $E_D$**

$$\frac{D_2}{D_1} = \frac{1.805 \times 10^{-11}}{4.826 \times 10^{-12}} = 3.7401575 \implies \ln(3.7401575) \approx 1.319128$$

$$\frac{1}{T_1} - \frac{1}{T_2} = \frac{1}{1123.15} - \frac{1}{1223.15} = 8.903530 \times 10^{-4} - 8.175612 \times 10^{-4} = 7.27918 \times 10^{-5}\ \mathrm{K}^{-1}$$

$$E_D = \frac{8.314 \times 1.319128}{7.27918 \times 10^{-5}} = 150\,666\ \mathrm{J/mol} = 150.67\ \mathrm{kJ/mol}$$

$$E_D = \frac{150\,666\ \mathrm{J/mol}}{4184\ \mathrm{J/kcal}} \approx \mathbf{36.01}\ \mathrm{kcal/mol} \approx \mathbf{36}\ \mathrm{kcal/mol}$$

**Part (b): Diffusivity at $1000^\circ\mathrm{C}$ ($T_3 = 1273.15$ K)**

$$\frac{1}{T_2} - \frac{1}{T_3} = \frac{1}{1223.15} - \frac{1}{1273.15} = 8.175612 \times 10^{-4} - 7.854534 \times 10^{-4} = 3.21078 \times 10^{-5}\ \mathrm{K}^{-1}$$

$$\text{Exponent} = \frac{150\,666}{8.314} \times (3.21078 \times 10^{-5}) = 18121.96 \times 3.21078 \times 10^{-5} \approx 0.581857$$

$$D_3 = (1.805 \times 10^{-11}\ \mathrm{m^2/s}) \times e^{0.581857} = (1.805 \times 10^{-11}) \times 1.78936 \approx \mathbf{3.23 \times 10^{-11}}\ \mathrm{m^2/s}$$

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

**Consistency with Official Data:** The official reference table [Session 5 Slide 23] lists $D(\mathrm{C\ in\ Fe\text{-}FCC}) \approx 3 \times 10^{-11}\ \mathrm{m^2/s}$ at $1000^\circ\mathrm{C}$, matching our derived $3.23 \times 10^{-11}\ \mathrm{m^2/s}$; both are of the same order, the slide value being a one-significant-figure figure ($\approx 8\%$ lower). **Mechanism Check:** $E_D \approx 36$ kcal/mol is lower than the activation energy for self-diffusion of iron ( $57.5$ kcal/mol $= 240$ kJ/mol for $\alpha$-Fe) [Slide 9]. This agrees with carbon diffusing interstitially, which needs less energy than vacancy diffusion [Slide 13].

Problem 4 | Problems T3_Diffusion.pdf #4 | Fick's 1st Law · Ceramic Barrier Transport

# Ionic Transport of Nickel Through an MgO Ceramic Plate

Original statement · Problems T3_Diffusion.pdf

*4.- A plate of magnesium oxide of* $0.1\ \text{cm}$ *thickness separates two metallic blocks, one of* $\text{Ni}$ *and one of* $\text{Ta}$*. At* $1400^\circ\text{C}$ $\text{Ni}$ *ions diffuse through the* $\text{MgO}$ *plate. The diffusion coefficient of* $\text{Ni}$ *in* $\text{MgO}$ *is* $9 \times 10^{-12}\ \text{cm}^2/\text{s}$ *and the lattice parameter of (fcc)* $\text{Ni}$ *at* $1400^\circ\text{C}$ *is* $3.6 \times 10^{-8}\ \text{cm}$*. Determine the time necessary for sufficient ions* $\text{Ni}^{2+}$ *ions to pass through the ceramic material so that the thickness of the block of* $\text{Ni}$ *is reduced by one micron.* Official Solution: t = 309 h

PHASE 1 · HYPOTHESES & PARAMETERS

- **Quasi-Steady State:** Because the plate thickness ($\Delta x = 0.1\ \text{cm} = 1000\ \mu\text{m}$) is much larger than the layer consumed ($\Delta h = 1\ \mu\text{m}$), flux through $\text{MgO}$ is in steady state [Slide 16].
- **Interfacial Boundary Conditions:** At the $\text{Ni}/\text{MgO}$ interface, the concentration equals the atomic density of nickel metal: $C_1 = \rho_{\text{at, Ni}} = 4/a^3$ (FCC, as the statement gives). At the $\text{MgO}/\text{Ta}$ interface, the nickel that arrives is taken as removed at once: $C_2 \approx 0$ (assumption of the model).
- **Parameters:** $\Delta x = 0.1\ \text{cm}$, $D = 9 \times 10^{-12}\ \text{cm}^2/\text{s}$, $a = 3.6 \times 10^{-8}\ \text{cm}$, $\Delta h = 1\ \mu\text{m} = 10^{-4}\ \text{cm}$.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

1. Atoms of Ni removed per unit area: $N/A = \rho_{\text{at, Ni}} \cdot \Delta h = \frac{4 \cdot \Delta h}{a^3}$.
2. Steady-state flux via Fick's First Law [Session 5 Slide 16]:

$$J = D\frac{\Delta C}{\Delta x} = D\frac{\rho_{\text{at, Ni}}}{\Delta x}$$

3. Conservation of mass equates flux to removal rate:

$$J = \frac{N/A}{t} \implies \frac{\rho_{\text{at, Ni}} \cdot \Delta h}{t} = D\frac{\rho_{\text{at, Ni}}}{\Delta x} \implies t = \frac{\Delta x \cdot \Delta h}{D}$$

The atomic density $\rho_{\text{at, Ni}}$ elegantly cancels out.

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

**1. Direct Evaluation with General Formula:**

$$\Delta x = 0.1\ \text{cm}, \quad \Delta h = 10^{-4}\ \text{cm}, \quad D = 9 \times 10^{-12}\ \text{cm}^2/\text{s}$$

$$t = \frac{(0.1\ \text{cm}) \times (10^{-4}\ \text{cm})}{9 \times 10^{-12}\ \text{cm}^2/\text{s}} = \frac{10^{-5}\ \text{cm}^2}{9 \times 10^{-12}\ \text{cm}^2/\text{s}} = \frac{1}{9} \times 10^{7}\ \text{s} \approx 1.1111 \times 10^{6}\ \text{s}$$

**2. Conversion to Hours:**

$$t = \frac{1.1111 \times 10^{6}\ \text{s}}{3600\ \text{s/h}} \approx \mathbf{308.64}\ \text{h} \approx \mathbf{309}\ \text{h}$$

**3. Step-by-Step Atomic Confirmation:**

$$\rho_{\text{at, Ni}} = \frac{4}{(3.6 \times 10^{-8}\ \text{cm})^3} = 8.5734 \times 10^{22}\ \text{atoms/cm}^3$$

$$N/A = (8.5734 \times 10^{22}) \times (10^{-4}) = 8.5734 \times 10^{18}\ \text{atoms/cm}^2$$

$$J = (9 \times 10^{-12}) \times \frac{8.5734 \times 10^{22}}{0.1} = 7.71605 \times 10^{12}\ \text{atoms/cm}^2 \cdot \text{s}$$

$$t = \frac{8.5734 \times 10^{18}}{7.71605 \times 10^{12}} = 1.1111 \times 10^{6}\ \text{s} \approx \mathbf{309}\ \text{h}$$

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

**Dimensional Check:** $[t] = \frac{\text{cm} \cdot \text{cm}}{\text{cm}^2/\text{s}} = \text{s}$. **Physical Insight:** Nearly 13 days (309 hours) at 1400°C are needed to remove 1 $\mu$m of nickel: diffusion is slower for charged species, in compact structures and in materials with high $T_m$ [Slide 27]. At fixed concentrations the flux falls as $1/\Delta x$, so a thicker plate would take proportionally longer [Slide 16].

Problem 5 | Problems T3_Diffusion.pdf #5 | Fick's 2nd Law + Arrhenius Coupled

# Cementation Temperature for Steel 1010 in 8 Hours

Original statement · Problems T3_Diffusion.pdf

*5.- A 1010 steel, that contains* $0.10\ \text{wt\%}\ \text{C}$ *is subjected to cementation process, in order to obtain effective* $\text{C}$ *composition of* $0.30\ \text{wt\%}$ *at a depth of* $1.16\ \text{mm}$ *below the surface. Assume that the maximum* $\text{C}$ *solubility in the steel surface is reached at moment that the carburizing process is initiated, generating a* $\text{C}$ *concentration at the surface of* $1.31\ \text{wt\%}$*. Determine the required cementation temperature for the process to be successfully carried out within an 8 hour time span. Data:* $D_0(\text{C in Fe}) = 16.2\ \text{mm}^2/\text{s}$*;* $E_D = 137\,800\ \text{J/mol}$*;* $R = 8.314\ \text{J/mol}\cdot\text{K}$.

**Official Solution: T = 1175 K = 902 °C**

PHASE 1 · HYPOTHESES & PARAMETERS

- **Coupled Transient Diffusion:** Combines semi-infinite error function diffusion with Arrhenius thermal kinetics [Session 5 Slides 20, 26].
- **Parameters:** $C_0 = 0.10\ \text{wt\%}$, $C_s = 1.31\ \text{wt\%}$, $C_x = 0.30\ \text{wt\%}$ $x = 1.16\ \text{mm} = 1.16 \times 10^{-3}\ \text{m}$ $t = 8\ \text{h} = 8 \times 3600\ \text{s} = 28\,800\ \text{s}$ $D_0 = 16.2\ \text{mm}^2/\text{s} = 16.2 \times 10^{-6}\ \text{m}^2/\text{s}$ $E_D = 137\,800\ \text{J/mol}$, $R = 8.314\ \text{J/mol}\cdot\text{K}$.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

1. Evaluate the concentration ratio $\text{erf}(z) = \frac{C_s - C_x}{C_s - C_0}$.
2. Determine $z$ via linear interpolation in the official table [Slide 21].
3. Calculate the required diffusivity: $D = \frac{x^2}{4 \cdot z^2 \cdot t}$.
4. Invert the Arrhenius equation to calculate the processing temperature $T$:

$$T = \frac{-E_D}{R \cdot \ln(D/D_0)}$$

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

**1. Error Function Value:**

$$\operatorname{erf}(z) = \frac{1.31 - 0.30}{1.31 - 0.10} = \frac{1.01}{1.21} \approx 0.83471074$$

**2. Linear Interpolation in Error Function Table [Slide 21]:**

- For $z_1 = 0.95 \implies \operatorname{erf}(z_1) = 0.8209$
- For $z_2 = 1.00 \implies \operatorname{erf}(z_2) = 0.8427$

$$\frac{z - 0.95}{1.00 - 0.95} = \frac{0.83471074 - 0.8209}{0.8427 - 0.8209} \implies \frac{z - 0.95}{0.05} = \frac{0.01381074}{0.0218} \approx 0.633520$$

$$z = 0.95 + 0.05 \times 0.633520 = 0.981676 \implies z^2 = (0.981676)^2 \approx 0.963688$$

**3. Diffusivity Calculation:**

$$x^2 = (1.16 \times 10^{-3}\ \mathrm{m})^2 = 1.3456 \times 10^{-6}\ \mathrm{m}^2$$

$$4 \cdot z^2 \cdot t = 4 \times 0.963688 \times 28\,800\ \mathrm{s} = 111\,016.86\ \mathrm{s}$$

$$D = \frac{1.3456 \times 10^{-6}\ \mathrm{m}^2}{111\,016.86\ \mathrm{s}} \approx \mathbf{1.21207 \times 10^{-11}}\ \mathrm{m^2/s}$$

**4. Inverting Arrhenius for Temperature $T$:**

$$\frac{D}{D_0} = \frac{1.21207 \times 10^{-11}\ \mathrm{m^2/s}}{16.2 \times 10^{-6}\ \mathrm{m^2/s}} = 7.48191 \times 10^{-7} \implies \ln\left(\frac{D}{D_0}\right) = \ln(7.48191 \times 10^{-7}) = 2.01249 - 16.11810 = -14.10561$$

$$T = \frac{-137\,800\ \mathrm{J/mol}}{(8.314\ \mathrm{J/mol \cdot K}) \times (-14.10561)} = \frac{137\,800}{(8.314)(14.10561)} = \frac{137\,800}{117.275} \approx \mathbf{1175.02}\ \mathrm{K} \approx \mathbf{1175}\ \mathrm{K}$$

$$T = 1175.02 - 273.15 = \mathbf{901.87}^{\circ}\mathrm{C} \approx \mathbf{902}^{\circ}\mathrm{C}$$

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

**Sensitivity to Temperature:** To reach $0.30\ \mathrm{wt}\%$ at $1.16\ \mathrm{mm}$ in $8\ \mathrm{h}$ the steel must be carburized at $902^{\circ}\mathrm{C}$. Because $D$ grows exponentially with $T$, a lower temperature would need a much longer time and a higher one a shorter time [Slide 26].

Problem 6 | Problems T3_Diffusion.pdf #6 | Fick's 1st Law · Membrane Purification

# Hydrogen Purification Through a Palladium Membrane

Original statement · Problems T3_Diffusion.pdf

*6.- Determine the thickness of a plate of* $\mathrm{Pd}$ *with transversal area of* $0.2\ \mathrm{m}^2$ *so that it can purify* $1.73 \times 10^{-3}\ \mathrm{kg/h}$ *of hydrogen, if the hydrogen concentration at the side of high pressure of the plate is* $1.5\ \mathrm{kg/m}^3$ *and at the side of low pressure is* $0.3\ \mathrm{kg/m}^3$*. The diffusion coefficient of hydrogen in* $\mathrm{Pd}$ *is* $1 \times 10^{-8}\ \mathrm{m}^2/\mathrm{s}$. Official Solution: Δx = 5 mm

PHASE 1 · HYPOTHESES & PARAMETERS

- **Pure Steady State:** Constant external pressures create time-invariant boundary concentrations $C_{\text{high}}$ and $C_{\text{low}}$ [Session 5 Slide 35].
- **Parameters:** $A = 0.2\ \mathrm{m}^2, \dot{m} = 1.73 \times 10^{-3}\ \mathrm{kg/h}\ C_1 = 1.5\ \mathrm{kg/m}^3, C_2 = 0.3\ \mathrm{kg/m}^3\ D = 1.0 \times 10^{-8}\ \mathrm{m}^2/\mathrm{s}$.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

1. Compute mass flux density [Session 5 Slide 15]:

$$J = \frac{\dot{m}}{A} \quad \left[\frac{\mathrm{kg}}{\mathrm{m}^2 \cdot \mathrm{s}}\right]$$

2. Apply Fick's First Law across the flat membrane [Slide 16, 35]:

$$J = D\frac{\Delta C}{\Delta x} \implies \Delta x = \frac{D \cdot (C_1 - C_2)}{J}$$

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

**1. Flux Density ($J$):**

$$\frac{\dot{m}}{A} = \frac{1.73 \times 10^{-3}\ \mathrm{kg/h}}{0.2\ \mathrm{m^2}} = 8.65 \times 10^{-3}\ \mathrm{kg/m^2 \cdot h}$$

$$J = \frac{8.65 \times 10^{-3}\ \mathrm{kg/m^2 \cdot h}}{3600\ \mathrm{s/h}} \approx \mathbf{2.4028 \times 10^{-6}}\ \mathrm{kg/m^2 \cdot s}$$

*The worked solution stored in the text layer of Official Slide 35 gives* $J = 8.65 \times 10^{-3}\ \mathrm{kg/m^2 \cdot h} = 2.4 \times 10^{-6}\ \mathrm{kg/m^2 \cdot s}$ *and* $\Delta x = 5\ \mathrm{mm}$ *(the answer box on the rendered slide appears empty).*

**2. Concentration Difference:**

$$\Delta C = C_1 - C_2 = 1.5\ \mathrm{kg/m^3} - 0.3\ \mathrm{kg/m^3} = 1.2\ \mathrm{kg/m^3}$$

**3. Membrane Thickness ($\Delta x$):**

$$\Delta x = \frac{(1.0 \times 10^{-8}\ \mathrm{m^2/s}) \times (1.2\ \mathrm{kg/m^3})}{2.4028 \times 10^{-6}\ \mathrm{kg/m^2 \cdot s}} \approx 4.994 \times 10^{-3}\ \mathrm{m} \approx \mathbf{5.0 \times 10^{-3}}\ \mathrm{m} = \mathbf{5}\ \mathrm{mm}$$

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

**Dimensional Check:** $[\Delta x] = \frac{(\mathrm{m^2/s}) \cdot (\mathrm{kg/m^3})}{\mathrm{kg/(m^2 \cdot s)}} = \mathrm{m}$. **Physical Insight:** At fixed concentrations the flux scales as $J \propto 1/\Delta x$ [Slide 16], so 5 mm is the thickness at which steady-state permeation equals the required purified flow. Hydrogen is small with respect to the matrix, so it diffuses interstitially, which is faster [Slide 13]; its diffusivity ($10^{-8}\ \mathrm{m^2/s}$) is far above the values of the diffusivity table [Slide 23]. This is the example of $\mathrm{H_2}$ purification through a $\mathrm{Pd}$ layer [Slide 35].